

Inheritance in java

inheritance is an important of concept of OOP

Inheritance is a process to make a father-child relationship through which one class acquires properties and behaviour(methods & attributes) of other class.

Process for achieving Inheritance

1. the information is made **manageable in a hierarchical order**. (from a father to its child)
2. The class which inherits the properties of other is known as **subclass** (derived class, child class)
3. the class whose properties are inherited is known as **superclass** (base class, parent class)
4. It can be achieved by **extends** keyword

This kind of relationship is also called **is-A** relationship

Let see some example

1. We have class calculator

```
public class Calculator {  
  
    public int add(int x, int y) {  
        return x + y;  
    }  
  
    public int add(int x, int y,int z) {  
        return x + y;  
    }  
  
    public int sub(int x, int y) {  
        return x - y;  
    }  
}
```

2. Suppose some one created this calculator class and we need to build a Advance calculator which will perform some more operations
3. If this Calculator is used by other programmers.(suppose if this calculator class object is used in multiple main method/Driver classes)
4. We will not try to directly change this Calculator class.
5. What we will do,we will extend the existing functionality of this class by inheriting it or extending it.

```
public class AdvCalculator extends Calculator {  
    public int mul(int x,int y){  
        return x*y;  
    }  
    public int div(int x,int y){  
        return x/y;  
    }  
}
```

- Now lets us make use of this Calculator class

```
public class Driver {  
    Run | Debug  
    public static void main(String[] args) {  
        int a = 4;  
        int b = 5;  
        Calculator cal = new Calculator();  
  
        int summation = cal.add(a, b);  
        int subtraction = cal.sub(a, b);  
        System.out.println("summation:" + summation);  
        System.out.println("subtraction:" + subtraction);  
    }  
}
```

1. The Calculator class has three methods add (2 parameters) ,add(3 parameters) and sub only.
2. AdvCalculator class has only two method mul and div but it will inherit other three methods add,add and sub from it's parent class Calculator

```

public class Driver {

    Run | Debug
    public static void main(String[] args) {
        int a = 4;
        int b = 5;
        int c = 6;
        AdvCalculator cal = new AdvCalculator();

        int summation = cal.add(a, b);
        int subtraction = cal.sub(a, b);
        System.out.println("summation:" + summation);
        System.out.println("subtraction:" + subtraction);

        int summation3 = cal.add(a, b, c);
        System.out.println("summation:" + summation3);

        int multiplication = cal.mul(a, b);
        System.out.println("multiplication:" + multiplication);
    }
}

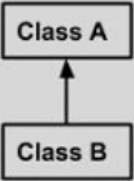
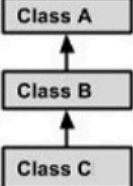
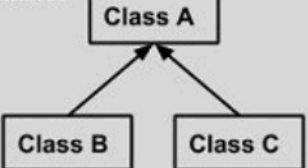
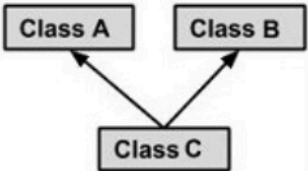
```

Let us Analyse

1. Calculator is **super/parent/base** class and AdvCalculator is **child/derived** class.
2. AdvCalculator **is-A** Calculator.
3. Calculator has 3 methods while AdvCalculator has 5 methods, 2 of its own and 3 inherited.
4. Properties can also be inherited.
5. Only private properties and methods are not inherited.

Types of Java Inheritance

1. There are 3 types of inheritances **Single**, **Multilevel**, and **Hierarchical** supported in java
2. Java does not support **Multiple** and **Hybrid** inheritance.

Single Inheritance  <pre>graph BT; B[Class B] --> A[Class A]</pre>	<pre>public class A { } public class B extends A { }</pre>
Multi Level Inheritance  <pre>graph BT; C[Class C] --> B[Class B]; B --> A[Class A]</pre>	<pre>public class A {} public class B extends A {.....} public class C extends B {.....}</pre>
Hierarchical Inheritance  <pre>graph BT; B[Class B] --> A[Class A]; C[Class C] --> A</pre>	<pre>public class A {} public class B extends A {.....} public class C extends A {.....}</pre>
Multiple Inheritance  <pre>graph BT; A[Class A] --> C[Class C]; B[Class B] --> C</pre>	<pre>public class A {} public class B {.....} public class C extends A,B { } // Java does not support mutiple Inheritance</pre>